HARAMAYA UNIVERSITY | DEPARTMENT OF COMPUTER SCIENCE

90-DAY JAVA MASTERY CHALLENGE WORKSHEET

CoSc3053: Java Programming | Daily Progress & Hands-On Engineering Log

Student Name: _____ **Student ID:** _____

Target Completion Date: _____ **Supervisor / Lecturer:** Lecturer Kidane W.

1. Day 1 Kickoff Challenge: Environment Setup & Diagnostic Engine

Welcome to Day 1 of your 90-Day Java Mastery Challenge! Today's goal is to establish a pristine development environment by installing JDK 25 LTS, IntelliJ IDEA (Community Edition), and Git, then verifying compilation and execution mechanics via command-line tools and IDE diagnostics.

🛠️ Step-by-Step Tooling Installation Guide

1. Install Temurin JDK 25 (LTS):
   - Download the Temurin 25 LTS installer.
   - Enable both 'Add to PATH' and 'Set JAVA_HOME variable' options during setup.

2. Install IntelliJ IDEA (Community Edition / Unified Free Core):
   - Download IntelliJ IDEA installer.
   - Select 'Create Desktop Shortcut', 'Add launcher dir to PATH', and '.java association'.

3. Install Git SCM:
   - Download Git installer.
   - Accept standard terminal integration and default branch options.

1.1 Command-Line Environment Verification

Terminal Verification Commands

```
$ java -version
openjdk version "25.0.1" 2026-09-15
OpenJDK Runtime Environment Temurin-25.0.1+8 (build 25.0.1+8)

$ javac -version
javac 25.0.1

$ git --version
git version 2.45.2.windows.1
```

1.2 Executable Java Programs: Hello World & System Environment Diagnostic

Program 1: HelloJava.java (Basic Entry Point)

```
package edu.haramaya.cosc3053.day1;

public class HelloJava {
    public static void main(String[] args) {
        System.out.println("Hello, Haramaya University!");
        System.out.println("Day 1 of 90-Day Java Challenge Successfully Started.");
    }
}
```

Program 2: EnvironmentDiagnostic.java (System Property Audit)

```
package edu.haramaya.cosc3053.day1;

public class EnvironmentDiagnostic {
    public static void main(String[] args) {
        System.out.println("========================================");
        System.out.println("   JAVA ENVIRONMENT DIAGNOSTIC TRACE    ");
        System.out.println("========================================");
        System.out.println(" Java Version : " + System.getProperty("java.version"));
        System.out.println(" Java Vendor  : " + System.getProperty("java.vendor"));
        System.out.println(" OS Name      : " + System.getProperty("os.name"));
        System.out.println(" OS Arch      : " + System.getProperty("os.arch"));
        System.out.println(" User Name    : " + System.getProperty("user.name"));
        System.out.println("========================================");
    }
}
```

Visual Console Snapshot: Expected Output Trace

```
========================================
   JAVA ENVIRONMENT DIAGNOSTIC TRACE
========================================
 Java Version : 25.0.1
 Java Vendor  : Eclipse Adoptium
 OS Name      : Windows 11 / Linux
 OS Arch      : amd64
 User Name    : HaramayaStudent
========================================
```